

# Protecting the Router Is Necessary but Not Sufficient: A Placebo-Controlled Measurement of Router Precision in Post-Training Quantization of Mixture-of-Experts Models

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## Abstract

[**TODO:** Write the abstract last. Target 150–200 words, one paragraph, structured as: question → design → headline number → placebo → attribution → limitation. Concrete numbers to cite: router protection cuts top-1 expert flips by 48.9% (OLMoE) and 41.5% (Qwen) at INT4 with disjoint bootstrap confidence intervals; a parameter-matched placebo reproduces the unprotected baseline to six decimal places in routing KL; of the routing flips caused by exactly one mechanism, the router’s own rounded weights account for 43.4% at INT8 but 62.2% at INT4 on OLMoE, so the dominant source of drift changes with precision.]

## 1 Introduction

[**TODO:** P1 — the problem. A Mixture-of-Experts layer replaces a dense feed-forward block with many experts and a small router that selects top- $k$  of them per token (Shazeer et al., 2017; Fedus et al., 2022). Routing is a discrete  $\arg \text{top-}k$ , so unlike a dense matrix multiply it can change its output *categorically* under a small perturbation. Post-training quantization perturbs every weight. Motivate why that asymmetry matters.]

[**TODO:** P2 — the cheap structural fix. Routers are a vanishing fraction of the parameters (Table 2), so excluding them from quantization costs almost nothing in memory. State the hypothesis in one sentence.]

[**TODO:** P3 — the live disagreement, and our honest positioning. Fu et al. (2025) already run experts at W4A4 with the router at W8A8, so router protection is their *baseline*, not our invention — say this plainly. Meanwhile Anonymous (2026a) argue that a full-precision router is insufficient, because quantization error in expert *outputs* still shifts the next layer’s router logits, and therefore quantize the router and correct it with rank-aware losses. Nobody has measured the size of either term under

a controlled single-variable design with no calibration. That measurement is what this paper contributes.]

[**TODO:** P4 — what we do. Three policies differing in exactly one respect, two architectures, three bit-widths, no calibration data anywhere, bootstrap confidence intervals over sequences, a parameter-matched placebo control, and an offline four-cell decomposition of the residual drift.]

[**TODO:** P5 — findings, referring to Figure 1. Router protection produces large and replicable reductions in routing drift with disjoint confidence intervals in all six model  $\times$  bit-width cells; the placebo reproduces the unprotected baseline, so the effect is router-specific; but a substantial residual remains, and the attribution shows its dominant source *changes* with precision. Frame the answer as “necessary but not sufficient”.]

[**TODO:** P6 — contributions. Suggested four bullets: (i) the first placebo-controlled measurement of router precision as an isolated variable in MoE PTQ, with bootstrap CIs, on two architectures and three bit-widths; (ii) an adjudication of the Fu et al. (2025) / Anonymous (2026a) disagreement, answered quantitatively rather than rhetorically; (iii) an offline four-cell attribution separating the router’s own rounded weights from upstream activation drift, requiring no additional GPU runs; (iv) an open, gated implementation in which every run asserts what it actually quantized.]

## 2 Background

### 2.1 Mixture-of-Experts routing

[**TODO:** Define an MoE layer. For hidden state  $x \in \mathbb{R}^d$ , a router  $W_r \in \mathbb{R}^{E \times d}$  produces logits  $z = W_r x$ ; the layer selects the top- $k$  experts by  $z$  and combines their outputs weighted by the renormalised softmax over the selected entries only. Give the equation. Stress that the selection is discrete and that  $W_r$  is tiny relative to the  $E$  expert

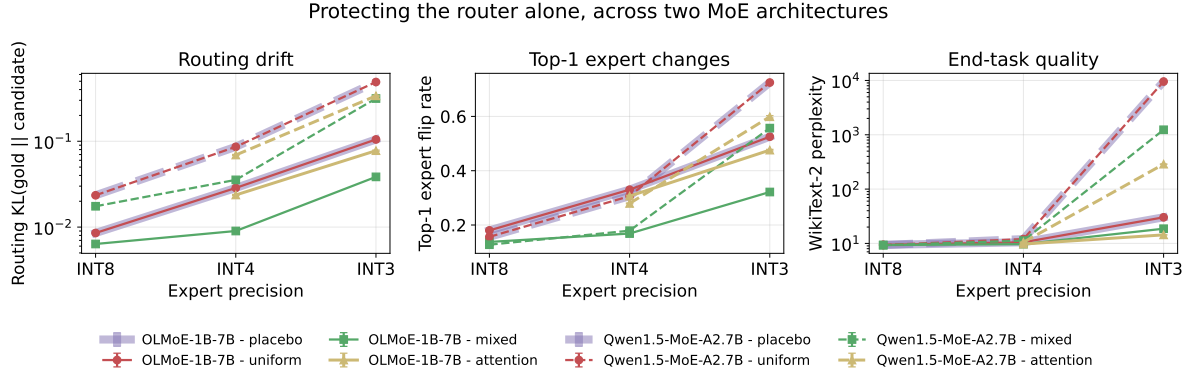


Figure 1: Routing drift (left, log scale), top-1 expert flip rate (centre) and WikiText-2 perplexity (right, log scale) against expert precision, for both architectures. MIXED keeps every router in BF16 while quantizing all experts; UNIFORM quantizes the routers too; PLACEBO protects a parameter-count-matched random non-router module instead of the routers; ATTENTION protects all attention projections instead of the routers and is run at INT4 and INT3 only. Error bars are 95% bootstrap confidence intervals over sequences on the two routing panels; perplexity is a point estimate over all scored tokens and carries none. The PLACEBO curve is drawn as a wide translucent underlay because it coincides with UNIFORM: that coincidence is the control result, and it is what makes the MIXED gap router-specific rather than a generic benefit of leaving some weights in high precision. The three panels do not agree, and that disagreement is the paper’s sharpest result: ATTENTION sits with UNIFORM on routing and with or below MIXED on perplexity (Section 6.3). [TODO: Check that the fonts in headline.pdf are legible at this width once compiled; the guidelines call this out explicitly. If not, widen the figure or regenerate with a larger base font size (see README.md).]

matrices. Cite Shazeer et al. (2017); Fedus et al. (2022); Jiang et al. (2024).]

[TODO: Define the *effective routing weights*: the post-top- $k$ , renormalised, zero-elsewhere vector that actually multiplies expert outputs. Explain why this differs from the full softmax, which is dominated by the long tail of experts that are never selected and can therefore move a great deal while the computation path is unchanged. We report both for that reason.]

## 2.2 Post-training quantization

[TODO: Define weight-only per-axis symmetric integer quantization: for each output channel,  $w \approx s \cdot q$  with  $q$  an  $N$ -bit integer and  $s$  a per-channel scale. State that this is the only quantizer used here, applied identically at 8, 4 and 3 bits via a single configuration class (Or et al., 2025), so that a trend across bit-widths reflects precision alone and not a change of kernel or algorithm. Contrast with calibration-based PTQ — GPTQ (Frantar et al., 2023), AWQ (Lin et al., 2024), SmoothQuant (Xiao et al., 2023), LLM.int8() (Dettmers et al., 2022) — and say that our question is deliberately about a *calibration-free structural* safeguard, which is why none of those appear as baselines.]

## 2.3 Metrics

[TODO: Define each metric in one or two sentences, in this order, and give the formula for the first three. **Routing KL**:  $\text{KL}(p_{\text{gold}} || p_{\text{cand}})$  over the full expert softmax, averaged over tokens and layers. **Top-1 flip rate**: the fraction of (token, layer) routing decisions whose  $\arg \max$  expert differs from gold. This is the headline metric because it counts decisions that actually changed. **Top- $k$  Jaccard distance**:  $1 - |S_{\text{gold}} \cap S_{\text{cand}}| / |S_{\text{gold}} \cup S_{\text{cand}}|$  over the selected expert sets; since both sets have size  $k$ , the union is  $2k$  minus the intersection. **Expert-usage entropy**: entropy of the marginal expert load, normalised by  $\log E$ ; the collapse detector. Note that we also report per-token routing entropy, because the marginal alone conflates load imbalance with router confidence. **Jensen-Shannon divergence**: reported alongside KL because KL is unbounded and explodes when the candidate places near-zero mass where gold had some. **Output-distribution drift**: a truncated KL over the gold model’s top- $M$  token ids ( $M = 64$ ) at every 16th scoring position, plus top-1 agreement — included because perplexity is blunt and a model can shift its output distribution noticeably while perplexity barely moves. **Perplexity**: WikiText-2, non-overlapping 1024-token windows.]

[TODO: Define the bootstrap. All intervals are 95% percentile bootstrap intervals with 200 resam-

ples, and *whole sequences are resampled, not individual tokens*. Explain why in one sentence: tokens within a sequence are correlated, so token-level re-sampling would produce dishonestly tight intervals. This matters because the entire Part 1 claim rests on intervals being disjoint.]

### 3 Related Work

**Router precision in MoE quantization.** [TODO: Fu et al. (2025) run experts at W4A4 with the router at W8A8 and benchmark OLMoE-1B-7B among others, so their numbers are directly comparable to our primary model. Router protection is their *baseline*, adopted without ablation. State the gap: its isolated contribution is never measured. Deng et al. (2026) independently report that routers are under 0.04% of MoE parameters — consistent with our own census (Table 2) — and fine-tune them; Anonymous (2026b) note that routers are typically left in full precision because they are small, yet errors from preceding layers still alter routing. [NOTE: sramoe is the one entry in custom.bib we could not verify — see its TODO there. Either confirm the OpenReview forum or drop this citation.]]

**Calibration-based routing repair.** [TODO: Park et al. (2026) align routing behaviour with a calibration objective combining value alignment and structure alignment at 4 bits; Hyeon and Do (2026) argue explicitly that retraining-free MoE compression is insufficient and that the router requires calibration. State the gap: these establish that calibration *helps*, but leave open how much of the problem a free structural choice already solves. Our design is the calibration-free lower bound against which such methods should be credited.]

**The disagreement we adjudicate.** [TODO: Anonymous (2026a) argue that freezing the router is *insufficient*, because quantization error in expert outputs still shifts the next layer’s router logits; they therefore quantize the router and correct it with a Rank-Aware Jaccard Loss and a Gap Hinge Loss. This is a mechanistic claim about *where* the residual drift comes from, and it is exactly what our four-cell attribution measures directly. Separately, Fang and Huang (2026) find that most routing errors are near-neighbour rank flips at the top- $k$  boundary — the strongest a priori reason to expect router protection to

underperform, and the failure mode that motivates the margin-aware top- $k$  widening we substitute for the proposal’s contingency plan (Section 5.6).]

**Bit-width sweeps over MoE sub-structures.** [TODO: Li et al. (2024) sweep bit-widths across MoE sub-structures and are our closest neighbour; Kim et al. (2023) find expert FFNs tolerate roughly 2-bit quantization while attention does not. Differentiate explicitly: neither isolates router precision as a single controlled variable, neither runs a parameter-matched placebo, and neither separates router-weight error from upstream activation drift. Say what we take from them (the sweep design, the expectation that experts are the robust part) and what we add.]

### 4 Method

#### 4.1 Three policies, one variable

[TODO: Define GOLD (nothing quantized; the BF16 upper bound and the reference every metric is computed against), UNIFORM (INT $N$  everywhere, routers included; the negative baseline), and MIXED (INT $N$  everywhere except the routers; the proposed safeguard). State the invariant that makes the comparison causal: UNIFORM and MIXED differ *only* in whether router modules are in the skip set — the symmetric difference of their skip sets contains nothing but router modules, and this is asserted in the test suite rather than assumed. Both policies additionally leave the token embedding and the output head in high precision, which is standard PTQ practice and identical across policies, so it cannot confound the comparison.]

#### 4.2 The placebo control

[TODO: This is the load-bearing control and deserves its own paragraph. Routers are 0.03% of OLMoE and 0.02% of Qwen (Table 2). If protecting *any* 0.02% of the weights bought the same improvement, the finding would be about high-precision weights in general, not about routers. PLACEBO therefore quantizes all routers, exactly as UNIFORM does, and instead protects a randomly chosen non-router module whose parameter count matches the total router budget to within 25%; the choice is seeded for reproducibility. Report what it selected: on both models the budget is met by a *single* expert projection matrix, so PLACEBO protects one module where MIXED protects 16 (OLMoE) or 24 (Qwen). [NOTE: This asymmetry is real and you must state it rather than let a reader discover it. It is

parameter-matched but not depth-matched. Carry the caveat into Section 8; do not bury it.]]

On both models the selection is a single expert down-projection in a single mid-depth layer: `model.layers.14.mlp.experts.10.down_proj` on OLMoE ( $1.000\times$  the router budget) and `model.layers.19.mlp.experts.45.down_proj` on Qwen ( $0.978\times$ ), both re-derived offline from the recorded `placebo_seed = 0` (Appendix B). Measured from the GOLD router logits, those modules are on the compute path for 11.4% (OLMoE) and 5.0% (Qwen) of routing tokens in one layer, against routers that every token reads in every layer — a token  $\times$  layer exposure gap of  $140\times$  and  $477\times$ .

### 4.3 The exposure-matched control

ATTENTION quantizes every router, exactly as UNIFORM does, and instead protects all attention projections: 64 modules and 268,435,456 parameters (3.88%) on OLMoE, 96 modules and 402,800,640 parameters (2.81%) on Qwen, or roughly  $128\times$  the router parameter budget. Attention is read by every token in every layer, which is the exposure profile the routers have and the PLACEBO does not. All 16 (resp. 24) routers remain quantized under ATTENTION, verified per router by name (Appendix A), so it is a genuine contrast with MIXED rather than a second MIXED run.

**[TODO: One paragraph on why this is the control PLACEBO cannot be. The two controls answer different questions: PLACEBO rules out “any equally-sized set of weights in BF16 helps”, ATTENTION rules out “any high-precision island on the every-token, every-layer main path helps”. Matched *exposure* rather than matched *parameter count* is the point, and being deliberately far larger than the router budget makes it the harder test on routing. Note that at nn.Linear granularity no policy can be matched on both count and exposure in either architecture — the smallest non-router linear layer in OLMoE is already  $16\times$  the per-layer router budget.]**

### 4.4 Four-cell attribution of the residual

**[TODO: Explain Table 1. Routing drift has exactly two possible sources. Cell A rebuilds gold from stored activations and weights; B applies quantized router weights to gold activations; C applies gold router weights to quantized activations, which is what MIXED does; D is both mechanisms at once. Every cell is scored against A, which makes B and**

| Activations | Router weights |            |
|-------------|----------------|------------|
|             | gold           | quantized  |
| gold        | A: reference   | B: weights |
| quantized   | C: = MIXED     | D: both    |

Table 1: The four-cell design. Routing drift has exactly two possible sources: rounding in the router’s own weights, and drift in the hidden states arriving at the router. Part 1 measures only their combination. Because router input activations are captured during the Part 1 runs, all four cells are single offline matrix multiplications,  $\text{logits} = XW^\top$ , and require no additional GPU jobs. All metrics are computed against cell A. Cell C is the MIXED condition, but cell D is **not** the UNIFORM run: it pairs quantized router weights with the MIXED run’s activations, so it omits the feedback of changed routing decisions on later layers’ hidden states. Table 15 measures that gap.

**C directly comparable. Be careful with D: it is UNIFORM *without routing feedback*, not UNIFORM itself, and the paper must not label it “= UNIFORM”. The gap between D and UNIFORM grows with damage — 50.01% vs 52.52% at OLMoE INT3 — and is worth one sentence as a measurement of the feedback term rather than being hidden.]**

**[TODO: Three implementation points that matter for correctness. (i) Activations for the quantized side come from the MIXED run, not UNIFORM: both quantize the same expert and attention weights, but only MIXED leaves the routers alone, so its hidden states isolate upstream drift from any feedback through changed routing decisions. (ii) The quantized router weights are read back from the UNIFORM run rather than re-derived offline, so the decomposition measures the rounding the experiment actually performed. (iii) The decomposition is gated on cell A reproducing the routing decisions the gold run recorded; the reconstruction runs in fp32 on CPU from fp16-stored activations while the model ran in BF16 on GPU, so the gate is on *decisions*, not bitwise values (Section 7).]**

**[TODO: State the non-additivity up front, before any numbers. Routing is an  $\arg\text{top-}k$  over a softmax, so the two mechanisms’ effects can cancel as easily as compound: a token that either source alone would have flipped is counted once in D but twice across B and C. We therefore report the interaction residual explicitly rather than folding it into either share, and we take as the interpretable quantity the *weight share of single-mechanism flips*,  $B/(B+C)$  — of the flips caused by exactly one mechanism, the fraction attributable to the router’s**



|                   | OLMoE-1B-7B   | Qwen1.5-MoE    |
|-------------------|---------------|----------------|
| Total parameters  | 6,919,161,856 | 14,315,784,192 |
| MoE layers        | 16            | 24             |
| Experts per layer | 64            | 60             |
| Top- $k$          | 8             | 4              |
| Hidden size       | 2,048         | 2,048          |
| Shared expert     | no            | yes            |
| Router parameters | 2,097,152     | 2,949,120      |
| Router share      | 0.0303%       | 0.0206%        |

Table 2: The two architectures, from scripts/inspect\_model.py. The router share is the quantity the whole hypothesis rests on: excluding the routers from quantization leaves three hundredths of one percent of the model in high precision. Checkpoints are allenai/OLMoE-1B-7B-0924 (Muennighoff et al., 2024) and Qwen/Qwen1.5-MoE-A2.7B (Qwen Team, 2024; Bai et al., 2023).

own weights. Above 50% means router protection is addressing the dominant source at that precision.]

## 5 Experimental Setup

### 5.1 Models

[**TODO:** One paragraph on the two models, referring to Table 2. Note that Qwen’s `shared_expert_gate` is deliberately *not* protected by MIXED: it emits a single scalar weighting the shared expert’s contribution rather than choosing among experts, so protecting it would make MIXED mean something broader on Qwen than on OLMoE, which has no shared expert. Leaving it quantized makes Qwen the conservative test, since whatever distortion it picks up counts against MIXED.]

### 5.2 Data and evaluation protocol

[**TODO:** WikiText-2 (Merity et al., 2017) test split, used for both perplexity and routing. Perplexity: non-overlapping 1024-token windows (window 1024, stride 1024) over the full split, so every token is scored exactly once — 288,448 scored tokens for OLMoE and 298,785 for Qwen, out of 288,730 and 299,078 tokenized tokens respectively. Routing metrics: 128 sequences of 256 tokens, seed 42, giving 32,768 routing decisions per layer, i.e. 524,288 (token, layer) decisions on OLMoE and 786,432 on Qwen. Single-sequence batches, so there is no padding and no masked positions to track. Router input activations are captured at every 8th token and stored as fp16 on CPU, giving 4,096 positions per layer for the attribution. State plainly that there is no training and no calibration set, so there is no train/test leakage surface: WikiText-2 is

used purely as an evaluation corpus and no model weight is fit to it.]

[**TODO:** Note the alignment guarantee, which is what licenses comparing gold and candidate logits row by row: batching is a pure function of (corpus, tokenizer, seed), and the gold run stores a SHA-256 fingerprint of the exact token ids. A candidate whose corpus, seed, tokenizer or sequence layout differs is rejected rather than silently compared against unrelated tokens.]

### 5.3 Quantization

[**TODO:** Weight-only per-axis (per-output-channel) symmetric integer quantization at 8, 4 and 3 bits via torchao’s `IntxWeightOnlyConfig` (Or et al., 2025), wired into transformers (Wolf et al., 2020) through `TorchAoConfig`. One configuration class for every bit-width, deliberately, so a trend across bit-widths reflects precision alone. We implement no quantization arithmetic; the contribution is the *policy* — which modules stay in high precision — and the verification that the policy was applied.]

### 5.4 Correctness gates

[**TODO:** Frame this correctly: every failure mode below is silent. A quantizer that skips a module it does not recognise still returns a working model with a plausible perplexity. The pipeline therefore asserts what it actually did rather than what it was asked to do, and `src/moequant/verify.py` runs after every load and refuses to proceed on failure. Enumerate the three traps: (i) **Routers that are not nn.Linear.** DeepSeek-style gates (Dai et al., 2024) hold a bare `nn.Parameter` and call `F.linear`, so a quantizer that only swaps `nn.Linear` never touches the router and UNIFORM silently becomes identical to MIXED — a vacuous comparison that still produces a full results table. Their forward also returns expert indices rather than logits, so a naive hook would compute KL over indices. (ii) **Fused expert stacks.** transformers 5.x fuses MoE experts into 3D parameter stacks that `nn.Linear`-based walkers skip, leaving the “quantized” experts at full precision while the load reports success; we pin transformers to 4.x and report the layout found at runtime. (iii) **Compute capability.** `LLM.int8()` requires capability 7.5+ (Dettmers et al., 2022) and BF16 is not native below Ampere, so the lab workstation’s cards are unusable and every run is gated on the allocated hardware.]

**[TODO:** Then list the assertions themselves and note that all of them passed on every run (Appendix A): GOLD quantizes nothing; every quantized policy quantizes something (catching a default rule that matched no module); MIXED has *no* router quantized; UNIFORM, PLACEBO and ATTENTION have *all* routers quantized; no module a policy protects was quantized anyway; each quantized weight shows at most  $2^N$  distinct values per output channel, checked independently of the library’s own bookkeeping and sampled so that a router and an expert are always inspected; and the config’s expert count equals the router weight’s output dimension. The GOLD run additionally compares its own captured logits against themselves and must yield exactly zero KL and zero flips — if it does not, something upstream is nondeterministic and no other number can be trusted.])

The single-variable claim is also checked *after the fact*, from the dequantized router weights each run stored, independently of the quantizer’s own bookkeeping. Across all 12 runs per model, every MIXED run’s router weights are bit-identical to GOLD in every layer, and every UNIFORM, PLACEBO and ATTENTION run’s routers sit at exactly the INT- $N$  level ceiling — 16 distinct values per output channel at INT4 and 8 at INT3, in all 1,024 (resp. 1,440) channels — with no layer matching GOLD (Table 9). This is a machine-checked result rather than an assertion about intent, and it is what licenses reading MIXED against UNIFORM as a one-variable contrast.

## 5.5 Hardware and reproducibility

**[TODO:** RTX 2080 Ti (compute capability 7.5), sharded with `device_map="auto"`: 3 cards for OLMoE (~14GB in BF16) and 5 for Qwen (~28.6GB). Software: Python 3.12.3, torch 2.9.1+cu128, transformers 4.57.6, torchao 0.17.0, datasets 5.0.1. Every results file records the software versions, GPU name and compute capability, and the git commit that produced it. Seed 42 throughout; 200 bootstrap resamples; the placebo module choice is seeded separately. State that quantization is deterministic, so the only randomness in the intervals is the sampling of sequences.])

## 5.6 Deviations from the approved proposal

**[TODO:** Write this as one honest paragraph or a short list, covering all four: (a) **Model substitution.** The proposal named Qwen/Qwen2.5-MoE-1.4B-A14B, which does

not exist on HuggingFace and is internally contradictory — 1.4B total parameters with 14B active is impossible. We use OLMoE-1B-7B as the primary model (small enough to iterate on, and benchmarked by Fu et al. (2025), so our numbers are comparable) and Qwen1.5-MoE-A2.7B as a second architecture with a structurally different router configuration. (b) **A bit-width sweep rather than INT8 alone.** At INT8 any effect risks being smaller than measurement noise, which would leave us unable to claim anything in either direction; a sweep also yields a trend rather than a single point. In the event this was the right call — the INT8 effect is real but the mechanism balance at INT8 is the *opposite* of that at INT4 (Section 7), which a single-point design would have missed entirely. (c) **Staged Part 1 / Part 2 framing**, so that a null Part 1 would still have produced a contribution. Part 1 was not null, and Part 2 became the mechanism analysis instead of a post-mortem. (d) **Contingency replaced.** The proposal’s fallback of dynamically scaling routing scores by activation size is replaced by margin-aware top- $k$  widening, which targets the near-neighbour rank-flip failure mode Fang and Huang (2026) identify. **[NOTE: Only claim this as future work — it is not implemented or evaluated here. Do not imply we ran it.]**

# 6 Results

## 6.1 Router protection helps, and the effect replicates

**[TODO:** One short paragraph. The claim is Tables 3 and 4: disjoint intervals in all six cells, on both metrics, on two architectures with different expert counts and different top- $k$ . Give the INT4 numbers as the headline (48.9% and 41.5% fewer top-1 flips) because INT4 is where the effect is largest without being saturated. Note that perplexity moves in the same direction throughout, so the routing result is not an artefact of a metric that nobody cares about.])

## 6.2 The effect is router-specific

**[TODO:** This is the paragraph the whole paper rests on, so give it room. Read the PLACEBO rows of Table 3 against the UNIFORM rows above them. At INT8 and INT4 the placebo’s routing KL is identical to UNIFORM’s to six decimal places on both models — 0.008535 vs 0.008535 and 0.028579 vs 0.028579 on OLMoE, 0.023470 vs 0.023470 and

| Policy                   | Bits | PPL       | Routing KL | Top-1 flip |
|--------------------------|------|-----------|------------|------------|
| <i>OLMoE-1B-7B</i>       |      |           |            |            |
| GOLD                     | BF16 | 8.3586    | 0.000000   | 0.00%      |
| UNIFORM                  | INT8 | 9.3941    | 0.008535   | 18.01%     |
| MIXED                    | INT8 | 9.3234    | 0.006356   | 13.76%     |
| PLACEBO                  | INT8 | 9.3941    | 0.008535   | 18.01%     |
| UNIFORM                  | INT4 | 10.4946   | 0.028579   | 33.01%     |
| MIXED                    | INT4 | 9.6938    | 0.009005   | 16.86%     |
| PLACEBO                  | INT4 | 10.4946   | 0.028579   | 33.01%     |
| ATTENTION                | INT4 | 9.7346    | 0.023662   | 30.65%     |
| UNIFORM                  | INT3 | 30.2290   | 0.104926   | 52.52%     |
| MIXED                    | INT3 | 18.6417   | 0.038463   | 32.15%     |
| PLACEBO                  | INT3 | 30.2259   | 0.104929   | 52.52%     |
| ATTENTION                | INT3 | 14.3592   | 0.077658   | 47.59%     |
| <i>Qwen1.5-MoE-A2.7B</i> |      |           |            |            |
| GOLD                     | BF16 | 7.9687    | 0.000000   | 0.00%      |
| UNIFORM                  | INT8 | 9.3465    | 0.023470   | 15.65%     |
| MIXED                    | INT8 | 9.2846    | 0.017484   | 12.75%     |
| PLACEBO                  | INT8 | 9.3461    | 0.023470   | 15.65%     |
| UNIFORM                  | INT4 | 12.0080   | 0.086117   | 30.62%     |
| MIXED                    | INT4 | 11.2445   | 0.035489   | 17.90%     |
| PLACEBO                  | INT4 | 12.0076   | 0.086117   | 30.62%     |
| ATTENTION                | INT4 | 10.5069   | 0.069109   | 27.95%     |
| UNIFORM                  | INT3 | 9555.4541 | 0.490062   | 72.48%     |
| MIXED                    | INT3 | 1236.4343 | 0.314730   | 55.69%     |
| PLACEBO                  | INT3 | 9563.0391 | 0.490072   | 72.48%     |
| ATTENTION                | INT3 | 287.7667  | 0.337966   | 59.87%     |

Table 3: Part 1 results. GOLD is the BF16 upper bound and the reference all routing metrics are computed against. The PLACEBO rows are the parameter-count-matched control and now cover all three bit-widths: at every bit-width the placebo reproduces UNIFORM across every metric, so the MIXED advantage is specific to the routers rather than a generic benefit of holding 0.02–0.03% of the weights in high precision. The ATTENTION rows are the exposure-matched control (Section 4.3), run at INT4 and INT3: it sits close to UNIFORM on the routing metrics while matching or beating MIXED on perplexity, which is the split Section 6.3 takes up. Top- $k$  Jaccard distance is reported in Appendix E and moves monotonically with the flip rate in every cell; pooled expert-usage entropy is omitted deliberately, because the per-layer analysis in Section 6.4 supersedes it.

0.086117 vs 0.086117 on Qwen — with top-1 flip rates equal to two decimal places. Protecting a parameter-count-matched random module instead of the routers does nothing, at any of the three bit-widths. Therefore the MIXED gap is not a generic consequence of holding some weights in high precision. [NOTE: Be precise about the INT3 placebo: there the agreement is close but not identical — routing KL 0.104929 vs 0.104926 on OLMoE and 0.490072 vs 0.490062 on Qwen, and perplexity 30.2259 vs 30.2290 and 9563.04 vs 9555.45. So write “identical to six decimal places at INT8 and INT4, and to four significant figures at INT3” rather than claiming six decimals everywhere. The direction of the tiny INT3 discrepancy is not even consistent between models, which is what you would expect from protecting one arbitrary module. Note also that the INT8 placebo agreement is not quite exact on Qwen either: perplexity 9.3461 vs 9.3465, with routing KL and flip rate identical to the reported precision.]]

[TODO: Then the limit of this control, stated here and not only in Limitations: it is matched on parameter count and nothing else, so it rules out “any equally-sized set of weights in BF16 helps” and cannot rule out “any high-precision island on the every-token, every-layer main path helps”. That is what ATTENTION is for.]

### 6.3 Routing and end-task quality come apart

On the routing metrics ATTENTION behaves like UNIFORM: at INT4 it recovers a quarter to a third of MIXED’s reduction in routing KL and a seventh to a fifth of its reduction in top-1 flips (Table 5), despite protecting roughly  $128\times$  the router parameter budget with the routers’ own exposure profile. On perplexity it ties MIXED once and beats it in the other three cells.

[TODO: This is the paper’s sharpest result and it needs the most careful writing in the paper. Land three things, in this order, from Table 5 and the three panels of Figure 1. (i) The routing verdict *supports* router specificity: protecting a hundred times more weight on the every-token, every-layer main path buys a fraction of what protecting the routers buys. (ii) The perplexity verdict contradicts any claim that routers are the best place to spend precision for end-task quality, and the paper must say so rather than hedge it. (iii) The reading that reconciles them: the two controls address different failure modes — protecting routers preserves *routing decisions*, protecting attention

| Model                                           | Bits | UNIFORM                    | MIXED                      | Reduction | Paired |
|-------------------------------------------------|------|----------------------------|----------------------------|-----------|--------|
| <i>Routing KL (<math>\times 10^{-3}</math>)</i> |      |                            |                            |           |        |
| OLMoE-1B-7B                                     | INT8 | 8.535 [8.326, 8.758]       | 6.356 [6.160, 6.550]       | 25.5%     | ✓      |
|                                                 | INT4 | 28.579 [28.096, 29.102]    | 9.005 [8.730, 9.304]       | 68.5%     | ✓      |
|                                                 | INT3 | 104.926 [102.413, 107.471] | 38.463 [36.767, 40.156]    | 63.3%     | ✓      |
| Qwen1.5-MoE-A2.7B                               | INT8 | 23.470 [22.859, 24.039]    | 17.484 [16.887, 18.043]    | 25.5%     | ✓      |
|                                                 | INT4 | 86.117 [84.862, 87.378]    | 35.489 [34.591, 36.461]    | 58.8%     | ✓      |
|                                                 | INT3 | 490.062 [484.581, 496.481] | 314.730 [310.617, 319.030] | 35.8%     | ✓      |
| <i>Top-1 flip rate (%)</i>                      |      |                            |                            |           |        |
| OLMoE-1B-7B                                     | INT8 | 18.01 [17.81, 18.18]       | 13.76 [13.61, 13.90]       | 23.6%     | ✓      |
|                                                 | INT4 | 33.01 [32.71, 33.31]       | 16.86 [16.71, 16.99]       | 48.9%     | ✓      |
|                                                 | INT3 | 52.52 [52.21, 52.83]       | 32.15 [31.86, 32.39]       | 38.8%     | ✓      |
| Qwen1.5-MoE-A2.7B                               | INT8 | 15.65 [15.47, 15.80]       | 12.75 [12.58, 12.90]       | 18.5%     | ✓      |
|                                                 | INT4 | 30.62 [30.44, 30.82]       | 17.90 [17.69, 18.13]       | 41.5%     | ✓      |
|                                                 | INT3 | 72.48 [72.11, 72.85]       | 55.69 [55.41, 55.99]       | 23.2%     | ✓      |

Table 4: Decision gate. **The intervals quoted throughout the paper are these:** 95% percentile bootstrap intervals over the full sample of 32,768 routing tokens per layer, resampling whole sequences, 200 resamples. In all six model  $\times$  bit-width cells the MIXED and UNIFORM intervals are disjoint on both metrics, so the advantage of router protection is larger than sampling noise everywhere we measured it. Comparing two independently bootstrapped intervals is a conservative stand-in for a paired-difference test rather than the test itself; the **Paired** column records that a paired-difference bootstrap with 10,000 replicates, computed offline on the strided capture, excludes zero in all twelve cells with 0.00% of replicates on the wrong side of it (Table 11). We quote the full-sample marginal intervals rather than the paired ones because the paired test can only be replayed at one token in eight, and no claim here depends on the choice.

preserves *output quality* — which is consistent with Kim et al. (2023), who find attention is the quantization-sensitive part while expert FFNs are robust. Routers stay overwhelmingly the more *efficient* place to spend precision (0.03% of parameters against 3.88%), which is a different claim from being the best one. Add that this is mechanistically coherent with Section 7 rather than in tension with it: ATTENTION removes part of the upstream activation drift that MIXED retains while leaving cell B’s router-weight error untouched, which predicts exactly the observed pattern of a modest routing gain and a real perplexity gain.]

Two caveats bound this. The Qwen INT3 row is inside the saturated regime of Section 6.5 — 9555, 1236 and 288 against a GOLD of 7.97 — and carries no weight in either direction. And the routing contrast is an INT4 statement: at INT3 ATTENTION recovers 41.0% (OLMoE) and 86.7% (Qwen) of MIXED’s routing-KL reduction, so the two policies separate least where the models are most damaged.

## 6.4 Expert collapse

Collapse occurs in exactly one cell of the sweep. On Qwen at INT3, UNIFORM starves 222 of 1440 expert slots and leaves an expert receiving zero tokens in five of 24 layers; MIXED cuts that to 55 slots with no fully unused expert anywhere, and the load distribution’s KL against uniform falls

$0.5717 \rightarrow 0.2693$ , a  $2.1\times$  reduction. PLACEBO reproduces UNIFORM here as elsewhere (220 slots, the same five layers). OLMoE never collapses at any bit-width.

**[TODO: One paragraph on Table 6, landing three things. (i) The pooled expert-usage entropy earlier drafts reported hid this: it is maximal at balance so its gradient is zero where the null sits, and it concatenates every layer’s tokens into one histogram, cancelling imbalances that point in different directions across layers — the same Qwen UNIFORM INT3 run moves only  $0.9996 \rightarrow 0.9929$  pooled while its worst layer falls to 0.7585 (Table 12). (ii) Router protection mitigates the collapse, and the placebo control passes on this metric too. (iii) OLMoE showing nothing is the honest boundary condition, not a failure: its unquantized baseline already starves 17 of 1024 slots, collapse appears only where a model is pushed past its breaking point, and OLMoE at INT3 is not there while Qwen is. Report the null as a result.]**

Both resolutions are stated because they differ: starved- and unused-expert counts come from the full 32,768 routing tokens per layer that each run recorded, while the load-distribution shape statistics are rebuilt from the strided capture at one token in eight, where low counts hit zero for reasons unrelated to quantization (Appendix D).



| Run                                             | UNIFORM | MIXED          | ATTENTION    | Rec.   |
|-------------------------------------------------|---------|----------------|--------------|--------|
| <i>Routing KL (<math>\times 10^{-3}</math>)</i> |         |                |              |        |
| OLMoE INT4                                      | 28.579  | <b>9.005</b>   | 23.662       | 25.1%  |
| OLMoE INT3                                      | 104.926 | <b>38.463</b>  | 77.658       | 41.0%  |
| Qwen INT4                                       | 86.117  | <b>35.489</b>  | 69.109       | 33.6%  |
| Qwen INT3                                       | 490.062 | <b>314.730</b> | 337.966      | 86.7%  |
| <i>Top-1 flip rate (%)</i>                      |         |                |              |        |
| OLMoE INT4                                      | 33.01   | <b>16.86</b>   | 30.65        | 14.6%  |
| OLMoE INT3                                      | 52.52   | <b>32.15</b>   | 47.59        | 24.2%  |
| Qwen INT4                                       | 30.62   | <b>17.90</b>   | 27.95        | 21.0%  |
| Qwen INT3                                       | 72.48   | <b>55.69</b>   | 59.87        | 75.1%  |
| <i>WikiText-2 perplexity</i>                    |         |                |              |        |
| OLMoE INT4                                      | 10.49   | <b>9.69</b>    | 9.73         | 94.9%  |
| OLMoE INT3                                      | 30.23   | 18.64          | <b>14.36</b> | 137.0% |
| Qwen INT4                                       | 12.01   | 11.24          | <b>10.51</b> | 196.6% |
| Qwen INT3                                       | 9555    | 1236           | <b>288</b>   | 111.4% |

Table 5: The exposure-matched control, and the split it produces. **Rec.** is the fraction of MIXED’s improvement over UNIFORM that ATTENTION recovers; above 100% means ATTENTION is better than MIXED. Best of the three in each block is bold. ATTENTION protects all attention projections — 3.88% of OLMoE and 2.81% of Qwen, about  $128\times$  the router parameter budget — and leaves every router quantized, with the routers’ own every-token, every-layer exposure. On the two routing metrics ATTENTION sits next to UNIFORM, recovering a quarter to a third of MIXED’s benefit at INT4 despite two orders of magnitude more protected weight; on perplexity it ties MIXED once and beats it in the other three cells. The same policy is therefore a fraction as effective as router protection at preserving routing decisions and at least as effective at preserving end-task quality; see Section 6.3. The two Qwen INT3 rows are inside the saturated regime of Section 6.5 and are also where the routing contrast is weakest. Recovered fractions are computed in `results/paper/comparison.md`.

## 6.5 Where the sweep stops being informative

[TODO: Say this before the discussion, not after. At INT3 on Qwen both policies have collapsed: perplexity 9555 for UNIFORM and 1236 for MIXED against a gold of 7.97. A 23% reduction in top-1 flips between two unusable models is not evidence for anything, and we do not count it. We report the cell for completeness and exclude it from every claim. OLMoE at INT3 (30.2 vs 18.6 against a gold of 8.36) is degraded but not destroyed, so we treat it as informative but weak. The load-bearing evidence is INT8 and INT4.]

## 7 Attribution Analysis

### 7.1 Reconstruction gate

[TODO: Report the gate before the findings, because nothing below is meaningful without it. Cell

| Run                         | UNIFORM       | MIXED         | PLACEBO | ATTENTION |
|-----------------------------|---------------|---------------|---------|-----------|
| <i>Starved expert slots</i> |               |               |         |           |
| OLMoE INT8                  | 13            | 13            | 13      | —         |
| OLMoE INT4                  | 15            | 15            | 15      | 12        |
| OLMoE INT3                  | 6             | 9             | 6       | 8         |
| Qwen INT8                   | 0             | 0             | 0       | —         |
| Qwen INT4                   | 1             | 0             | 1       | 0         |
| Qwen INT3                   | <b>222</b>    | <b>55</b>     | 220     | 74        |
| <i>KL(load    uniform)</i>  |               |               |         |           |
| OLMoE INT8                  | 0.1289        | 0.1292        | 0.1289  | —         |
| OLMoE INT4                  | 0.1291        | 0.1302        | 0.1291  | 0.1263    |
| OLMoE INT3                  | 0.1630        | 0.1405        | 0.1630  | 0.1419    |
| Qwen INT8                   | 0.0558        | 0.0547        | 0.0558  | —         |
| Qwen INT4                   | 0.0752        | 0.0644        | 0.0752  | 0.0653    |
| Qwen INT3                   | <b>0.5717</b> | <b>0.2693</b> | 0.5717  | 0.3461    |

Table 6: Expert collapse, resolved per layer. OLMoE has 1,024 (layer, expert) slots and Qwen 1,440; the unquantized GOLD baselines are 17 and 0 starved slots and load KL 0.1307 and 0.0465. A slot is *starved* when an expert receives less than a tenth of its fair share of tokens in its own layer; counts are summed over layers against the total number of (layer, expert) slots, since expert  $j$  in one layer is a different expert from expert  $j$  in another. Starved counts use all 32,768 routing tokens per layer; the load-distribution KL against the uniform load is rebuilt from the strided capture, where GOLD is measured on the same positions. Collapse appears in exactly one cell of the sweep, Qwen INT3, and router protection mitigates it: 222 starved slots and five layers containing an expert that receives zero tokens under UNIFORM, against 55 slots and no fully unused expert anywhere under MIXED. PLACEBO reproduces UNIFORM here as it does everywhere else. OLMoE never collapses — its unquantized baseline already starves 17 of 1024 slots and no policy is materially worse — which is the boundary condition, reported as a result. Appendix D has the per-layer detail and the pooled expert-usage entropy that averages all of this away.

A, rebuilt offline in fp32 on CPU from fp16-stored activations, reproduces the routing decisions the GPU BF16 gold run recorded at 99.66% mean agreement and 99.46% in the worst layer on OLMoE, and 99.37% / 98.49% on Qwen, against a required threshold of 95% (Table 16). Explain why the gate is on decisions and not values: the reconstruction is a different numerical path, so bitwise agreement is not available, but a decomposition of *routing decisions* only needs the decisions to match. Cross-check the rebuilt cells against the Part 1 runs they should reproduce (Table 15): cell C tracks MIXED to within a few tenths of a percentage point, which is the reconstruction gap from Part 1 measuring every token while the cells use the strided capture subset. Cell D sits *below* UNIFORM by a margin that grows with damage — 17.54 vs 18.01

| Model             | Bits | Top-1 flip rate vs. cell A (%) |                |         | $B/(B+C)$ (%)     | Interaction (%)      |
|-------------------|------|--------------------------------|----------------|---------|-------------------|----------------------|
|                   |      | B: weights                     | C: activations | D: both |                   |                      |
| OLMoE-1B-7B       | INT8 | 10.38                          | 13.54          | 17.54   | 43.4 [42.7, 44.1] | −36.4 [−38.0, −34.9] |
|                   | INT4 | 27.40                          | 16.62          | 31.41   | 62.2 [61.7, 62.8] | −40.2 [−41.4, −38.9] |
|                   | INT3 | 43.78                          | 31.74          | 50.01   | 58.0 [57.6, 58.4] | −51.0 [−52.1, −49.9] |
| Qwen1.5-MoE-A2.7B | INT8 | 8.81                           | 13.04          | 15.70   | 40.3 [39.8, 40.9] | −39.1 [−40.5, −37.7] |
|                   | INT4 | 25.00                          | 18.11          | 30.25   | 58.0 [57.5, 58.5] | −42.5 [−43.5, −41.6] |
|                   | INT3 | 45.76                          | 55.06          | 68.58   | 45.4 [45.1, 45.6] | −47.0 [−47.8, −46.2] |

Table 7: Part 2 attribution, 10,000 bootstrap replicates. Cells B, C and D follow Table 1 and are scored against the rebuilt reference cell A; cell D is both mechanisms at once, which is UNIFORM *without* the feedback of changed routing on later layers’ hidden states rather than UNIFORM itself.  $B/(B+C)$  is the interpretable column: of the routing flips caused by exactly one mechanism, the fraction attributable to the router’s own rounded weights, with 95% bootstrap intervals over sequences. Above 50% means router protection addresses the dominant source at that precision. The two mechanisms are *not* additive — a token that either source alone would flip is counted once in D but twice across B and C — so the interaction residual  $1 - B/D - C/D$  is reported explicitly, with its own interval, rather than folded into a share. Because it is large and negative everywhere, the shares of D do **not** partition the drift and must not be quoted as “ $X\%$  router weights,  $(100 - X)\%$  activation drift”; its negative sign means the two mechanisms flip overlapping sets of tokens. Figure 4 plots the same cells. **[NOTE: The Qwen INT3 row sits in the saturated regime dismissed in Section 6.5. It is the one cell where the share falls below half, and it must not be used as evidence — see the text.]**

at OLMoE INT8, 50.01 vs 52.52 at OLMoE INT3, 68.58 vs 72.48 at Qwen INT3 — because D omits the feedback of changed routing decisions on later layers’ hidden states. Report that gap as a measurement of the feedback term, and do not label D as UNIFORM.]

## 7.2 The dominant mechanism changes with precision

**[TODO: The core result, from Table 7 and Figure 4. At INT8 the router’s own weights account for a minority of single-mechanism flips — 43.4% on OLMoE and 40.3% on Qwen — so most of the drift that router protection cannot touch comes from hidden states that were already damaged upstream. At INT4 the ordering reverses: 62.2% and 58.0%, so the router’s own rounding becomes the larger single source, and protecting it is addressing the dominant term. The intervals are tight and the reversal is consistent across both architectures. Say what this means: the answer to “is protecting the router enough?” is precision-dependent, which is why the two positions in the literature can both be partly right.]**

**[NOTE: Be precise about what  $B/(B+C)$  does and does not say, because it is easy to garble. The residual that survives router protection is cell C by construction — MIXED is cell C, so its remaining drift is upstream activation drift, at every precision. What changes between INT8 and INT4 is not the composition of that residual but which of the two**

**mechanisms is larger. So the correct sentence is “at INT4 router protection removes the larger of the two terms, at INT8 it removes the smaller one”, not “at INT4 the residual is dominated by router weights”. The latter is false: after protection there are no router-weight errors left to dominate anything.]**

**[TODO: Then the honesty paragraph, and do not omit it. OLMoE at INT3 gives 58.0%, consistent with INT4. Qwen at INT3 gives 45.4%, the only cell below half — but that is the same saturated regime we excluded in Section 6.5, where both policies are destroyed. We therefore do not read it as a return of activation dominance at very low precision; it carries no argumentative weight in either direction. Make this explicit, because dismissing Qwen INT3 in Part 1 and then quoting it in Part 2 would be exactly the kind of selective reading this design was built to avoid.]**

**[TODO: Optionally, one sentence on the raw routing-KL split (Table 14), which shows the same crossover on a continuous metric: on OLMoE the weights-only KL is 0.001833 against 0.007284 for activations at INT8, and 0.017479 against 0.009920 at INT4. The crossover is therefore not an artefact of thresholding a continuous drift into a discrete flip.]**

## 8 Discussion

**Necessary but not sufficient. [TODO: Answer the research question in one sentence, then unpack**

it. Protecting the router is worth doing — the gains are large, replicable across architectures, placebo-controlled, and free — and it is not enough, because a substantial residual survives and at INT8 that residual is dominated by a mechanism router protection cannot reach.]

**Adjudicating the disagreement.** [TODO: Anonymous (2026a) argue full-precision routers are insufficient because expert-output error shifts downstream router logits; Fu et al. (2025) treat protection as a sensible default. Our numbers say both are describing real terms whose relative size depends on precision: the activation-drift term they emphasise dominates at INT8, while the router’s own rounding dominates at INT4. Frame our contribution as *measuring the size of both terms* under a controlled design, not as beating either.]

**Practical reading.** [TODO: What should a practitioner do? Router protection is free and always helped in our measurements, so it should be the default — but a method that only protects the router will hit a floor set by upstream drift, and that floor is highest (relatively) at the precisions where quantization is otherwise nearly lossless. Note that the calibration methods we cite (Park et al., 2026; Hyeon and Do, 2026) should be credited against this calibration-free floor rather than against a fully uniform baseline.]

**Why the mechanisms are not additive.** [TODO: One short paragraph on the consistently negative interaction residual (−36% to −51%): the two mechanisms flip overlapping sets of tokens, which is what you expect if flips concentrate on tokens whose top- $k$  margin was already narrow. Connect to Fang and Huang (2026)’s near-neighbour rank-flip finding — our residual is consistent with theirs, though we do not measure margins directly. [NOTE: Be careful: this is a consistency observation, not a measurement. We did not compute top- $k$  margins. Phrase it as a hypothesis the residual is compatible with.]]

## Limitations

[TODO: Write each of the following as one or two sentences. Do not soften them. (1) **The placebo is matched on parameter count and on nothing else.** On both models the router budget is met by a single non-router projection matrix, so PLACEBO

protects one module in one layer while MIXED protects 16 or 24 modules spread across every layer. The gap that matters is not depth but *activation exposure*: measured from the GOLD router logits, the protected module is on the compute path for 11.4% (OLMoE) and 5.0% (Qwen) of routing tokens in one layer, against routers read by every token in every layer — a token  $\times$  layer exposure gap of  $140\times$  and  $477\times$ . The placebo therefore rules out “any equally-sized set of weights in BF16 helps” and cannot rule out “any high-precision island on the every-token, every-layer main path helps”. It is also  $n = 1$ : one module drawn once, at placebo\_seed = 0. At nn.Linear granularity no control can be matched on count and exposure at once in either architecture, since the smallest non-router linear layer is already  $16\times$  the per-layer router budget; closing that would require quantizing row slices of a module, which the quantizer cannot express. (1b) **What ATTENTION does and does not settle.** It closes the exposure confound for *routing* but not for *perplexity*, where it matches or beats MIXED (Section 6.3): we can claim routers are the most efficient place to spend precision per protected parameter, not the best place for end-task quality. ATTENTION is also not parameter-matched, so it separates exposure from role only if protecting *more* weight on the main path cannot help *less*, and we have no policy matched on both count and exposure to check that. (2) **One corpus, one language, one task.** WikiText-2 perplexity and routing only; no downstream task evaluation, so we cannot say whether the routing changes we measure translate into task degradation. (3) **Weight-only quantization.** No activation quantization, so our setting is not directly comparable to W4A4 results such as Fu et al. (2025)’s. (4) **No calibration-based baseline.** By design — the question is about a calibration-free structural choice — but it means we cannot say how large the remaining headroom is relative to GPTQ-style methods. (5) **No memory or latency measurement.** torchao quantizes for real, but we run on GPUs where the dequantized compute path dominates and we measure quality, not throughput. Do not imply a speedup. (6) **Saturation at INT3.** Qwen INT3 is unusable under both policies, so one sixth of the sweep carries no evidential weight, in Part 1 or Part 2. (7) **The attribution uses a strided subset** (every 8th token, 4,096 positions per layer) and is reconstructed offline in a different numerical path from the run; the reconstruction gate bounds but does not eliminate that gap. The same stride

limits the paired-difference bootstrap and the load-shape statistics in Table 6, which is why the intervals we quote are the full-sample marginal ones (Table 4) and why starved-expert counts are taken from the full-resolution fields. **(8) Two architectures is not a survey.** Both have hidden size 2048 and both use a clean `nn.Linear` router; we did not evaluate a DeepSeek-style `nn.Parameter` gate, only defended against it. **(9) Perplexity carries no interval anywhere.** Per-window negative log-likelihood is not retained, so the first column of every results table is a point estimate over all 288,448 (OLMoE) / 298,785 (Qwen) scored tokens. Every inferential claim rests on the routing metrics, which do carry sequence-level intervals — and this is a real limit on Section 6.3, where the ATTENTION-versus-MIXED perplexity comparison is therefore a comparison of two point estimates.]

## 9 Conclusion

**[TODO:** Three to five sentences. Restate the question, the answer (“necessary but not sufficient”), the two numbers that carry it (placebo-controlled reduction at INT4; the 43%→62% shift in weight share between INT8 and INT4), and one forward pointer — margin-aware top- $k$  widening, targeting the near-neighbour rank-flip failure mode, as the natural next experiment. Do not introduce anything new here.]

## AI Disclosure and Reflection

**[TODO:** Write this yourselves — it is the one section where the content has to be your own record. Cover, per the guidelines: *how* and *why* you used AI; *which parts* of the project (e.g. implementation of the harness, the verification and correctness-gate design, plotting, literature search, LaTeX scaffolding, prose editing); *which tools and models* by name and version; and a genuine reflection on how it went, including where it was wrong or unhelpful.]

**[TODO:** Two concrete things worth reporting, because they are honest and specific. (i) An AI-assisted literature pass initially *conflated* two distinct papers — the anonymous “Beyond Freezing the Router” submission (Anonymous, 2026a) and Fang and Huang’s withdrawn “Router Choice Matters” (Fang and Huang, 2026) — and asserted they were the same work in our own project notes. It was caught by checking the sources directly, and the corrected distinction is now recorded as a comment

in `custom.bib` so it cannot be reintroduced. This is a useful, concrete example of why every citation was verified against the live source. (ii) The correctness gates in `verify.py` exist because the silent-failure modes in Section 5.4 are exactly the kind of bug that AI-assisted code produces confidently — a quantizer that walks `nn.Linear` modules looks correct and silently skips a `nn.Parameter` router. Reflect on the general lesson: the useful discipline was making the pipeline assert what it actually did.]

## References

- Anonymous. 2026a. [Beyond freezing the router: Rank-aligned post-training quantization for mixture-of-experts models](#). OpenReview. Anonymous submission under review.
- Anonymous. 2026b. Output-aware selective router alignment for MoE quantization. OpenReview. Anonymous submission under review; identifier not yet verified.
- Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. 2023. [Qwen technical report](#). *Preprint*, arXiv:2309.16609.
- Damai Dai, Chengqi Deng, Chenggang Zhao, R. X. Xu, Huazuo Gao, Deli Chen, Jiaoshi Li, Wangding Zeng, et al. 2024. [DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models](#). *Preprint*, arXiv:2401.06066.
- Jianing Deng, Song Wang, Dongwei Wang, Zijie Liu, Tianlong Chen, Huanrui Yang, and Jingtong Hu. 2026. [GEMQ: Global expert-level mixed-precision quantization for MoE LLMs](#). *Preprint*, arXiv:2605.23078.
- Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. 2022. [LLM.int8\(\): 8-bit matrix multiplication for transformers at scale](#). In *Advances in Neural Information Processing Systems*.
- Yi-Zeng Fang and Juinn-Dar Huang. 2026. [Router choice matters: Rank-aware post-training quantization for MoE models](#). OpenReview. Withdrawn submission to the International Conference on Learning Representations (ICLR) 2026.
- William Fedus, Barret Zoph, and Noam Shazeer. 2022. [Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity](#). *Journal of Machine Learning Research*, 23(120):1–39.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. 2023. [GPTQ: Accurate post-training quantization for generative pre-trained transformers](#). In *The Eleventh International Conference on Learning Representations*.



Zhongqian Fu, Tianyi Zhao, Ning Ding, Xianzhi Yu, Xiaosong Li, Yehui Tang, and Yunhe Wang. 2025. [EAQuant: Enhancing post-training quantization for MoE models via expert-aware optimization](#). *Preprint*, arXiv:2506.13329.

Sieun Hyeon and Jaeyoung Do. 2026. [Is retraining-free enough? the necessity of router calibration for efficient MoE compression](#). *Preprint*, arXiv:2603.02217.

Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, et al. 2024. [Mixtral of experts](#). *Preprint*, arXiv:2401.04088.

Young Jin Kim, Raffy Fahim, and Hany Hassan Awadalla. 2023. [Mixture of quantized experts \(MoQE\): Complementary effect of low-bit quantization and robustness](#). *Preprint*, arXiv:2310.02410.

Pingzhi Li, Xiaolong Jin, Zhen Tan, Yu Cheng, and Tianlong Chen. 2024. [QuantMoE-Bench: Examining post-training quantization for mixture-of-experts](#). *Preprint*, arXiv:2406.08155.

Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. 2024. [AWQ: Activation-aware weight quantization for LLM compression and acceleration](#). In *Proceedings of Machine Learning and Systems (MLSys)*.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. [Pointer sentinel mixture models](#). In *International Conference on Learning Representations*.

Niklas Muennighoff, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Jacob Morrison, Sewon Min, Weijia Shi, Pete Walsh, Oyvind Tafjord, Nathan Lambert, Yuling Gu, Shane Arora, Akshita Bhagia, Dustin Schwenk, David Wadden, Alexander Wettig, Binyuan Hui, Tim Dettmers, Douwe Kiela, and 5 others. 2024. [OLMoE: Open mixture-of-experts language models](#). *Preprint*, arXiv:2409.02060.

Andrew Or, Apurva Jain, Daniel Vega-Myhre, Jesse Cai, Charles David Hernandez, Zhenrui Zheng, Driss Guessous, Vasilii Kuznetsov, et al. 2025. [TorchAO: PyTorch-native training-to-serving model optimization](#). *Preprint*, arXiv:2507.16099.

Hancheol Park, Geonho Lee, Tairen Piao, and Tae-Ho Kim. 2026. [Value-and-structure alignment for routing-consistent quantization of mixture-of-experts models](#). *Preprint*, arXiv:2606.05688.

Qwen Team. 2024. [Qwen1.5-MoE: Matching 7b model performance with 1/3 activated parameters](#). Qwen blog.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. 2017. [Outrageously large neural networks: The sparsely-gated mixture-of-experts layer](#). In *International Conference on Learning Representations*.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, et al. 2020. [Transformers: State-of-the-art natural language processing](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45.

Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. 2023. [SmoothQuant: Accurate and efficient post-training quantization for large language models](#). In *Proceedings of the 40th International Conference on Machine Learning*.

## A Correctness gate report

| Model | Policy    | Quantized | Routers q. |
|-------|-----------|-----------|------------|
| OLMoE | GOLD      | 0         | 0 / 16     |
|       | ATTENTION | 3088      | 16 / 16    |
|       | MIXED     | 3136      | 0 / 16     |
|       | PLACEBO   | 3151      | 16 / 16    |
|       | UNIFORM   | 3152      | 16 / 16    |
| Qwen  | GOLD      | 0         | 0 / 24     |
|       | ATTENTION | 4440      | 24 / 24    |
|       | MIXED     | 4512      | 0 / 24     |
|       | PLACEBO   | 4535      | 24 / 24    |
|       | UNIFORM   | 4536      | 24 / 24    |

Table 8: Modules quantized per policy, identical at every bit-width. The counts are the proof that the policies differ as intended: MIXED quantizes exactly 16 (resp. 24) fewer modules than UNIFORM, and those modules are exactly the routers. PLACEBO quantizes all routers and spares one module instead, which is why its count is one below UNIFORM’s. ATTENTION spares 64 (resp. 96) attention projections and keeps every router quantized, which is what distinguishes it from MIXED. Every run also passed the per-channel distinct-value check ( $\leq 2^N$  levels) and the expert-count cross-check, and GOLD passed its self-comparison at exactly zero KL and zero flips. The module PLACEBO protected is not recorded in the results tree; it is deterministic given `placebo_seed = 0` and is re-derived in Table 10.

## B Offline verification of the policies

## C Paired-difference bootstrap

## D Expert collapse: per-layer detail

## E Part 1: additional metrics

**[TODO: Report normalised expert-usage entropy here, and do not describe it as showing an absence of expert collapse — Section 6.4 and Table 12 show that the pooled version hides a real collapse on Qwen INT3. Report it as the metric the earlier analysis relied on and explain why it is the wrong aggregate]**

| Model | Routers | Levels | Max | Identical |
|-------|---------|--------|-----|-----------|
| OLMoE | BF16    | 1161   | —   | 16 / 16   |
|       | INT8    | 60     | 256 | 0 / 16    |
|       | INT4    | 16     | 16  | 0 / 16    |
|       | INT3    | 8      | 8   | 0 / 16    |
|       |         |        |     |           |
| Qwen  | BF16    | 1107   | —   | 24 / 24   |
|       | INT8    | 46     | 256 | 0 / 24    |
|       | INT4    | 16     | 16  | 0 / 24    |
|       | INT3    | 8      | 8   | 0 / 24    |

Table 9: Machine-checked bit-identity and quantization level check, computed with no GPU from the dequantized router weights each run stored — that is, the weights the run actually used — rather than from the quantizer’s bookkeeping. The BF16 rows are GOLD and all three MIXED runs; the INT- $N$  rows are the UNIFORM and PLACEBO runs at that width, plus ATTENTION at INT4 and INT3. **Levels** is the largest number of distinct values found in any single output channel across all 1,024 (resp. 1,440) channels; per-axis symmetric INT- $N$  quantization cannot exceed the **Max** column,  $2^N$ . **Identical** counts layers whose router weights match GOLD bitwise. All 12 runs per model pass. The INT8 rows show fewer than 256 levels because with a per-channel scale of  $\max |w|/127$  and roughly Gaussian router weights the extreme codes are never reached; that is expected, not a shortfall. Regenerate with `scripts/verify_offline.py`; the full report is `results/<model>/verification.md`.

|                    | OLMoE     | Qwen      |
|--------------------|-----------|-----------|
| Layer / expert     | 14 / 10   | 19 / 45   |
| Parameters         | 2,097,152 | 2,883,584 |
| Ratio to budget    | 1.000×    | 0.978×    |
| Candidate pool     | 3,072     | 4,344     |
| Tokens on its path | 11.43%    | 5.04%     |
| Exposure gap       | 140×      | 477×      |

Table 10: What PLACEBO protected, re-derived offline from a meta-device skeleton at `placebo_seed = 0`: `model.layers.14.mlp.experts.10.down_proj` on OLMoE and `model.layers.19.mlp.experts.45.down_proj` on Qwen. Both are a *single* expert down-projection. The selection rule accumulates modules until within 25% of the router budget, but on both real checkpoints every eligible candidate is already within tolerance on its own, so the loop terminates after one draw and the tolerance band is never exercised. **Tokens on its path** is measured from the GOLD router logits over the 32,768 routing tokens; the exposure gap counts token  $\times$  layer forward events against routers read by every token in every layer.

| Bits         | Metric | Diff.    | 95% paired CI        |
|--------------|--------|----------|----------------------|
| <i>OLMoE</i> |        |          |                      |
| INT8         | KL     | −2.123   | [−2.178, −2.071]     |
| INT8         | flips  | −41.901  | [−44.357, −39.413]   |
| INT4         | KL     | −19.654  | [−19.996, −19.350]   |
| INT4         | flips  | −156.982 | [−161.728, −152.435] |
| INT3         | KL     | −67.093  | [−69.430, −65.174]   |
| INT3         | flips  | −201.645 | [−206.863, −196.594] |
| <i>Qwen</i>  |        |          |                      |
| INT8         | KL     | −5.809   | [−5.966, −5.660]     |
| INT8         | flips  | −28.778  | [−30.609, −26.978]   |
| INT4         | KL     | −50.866  | [−51.526, −50.205]   |
| INT4         | flips  | −129.242 | [−132.833, −125.661] |
| INT3         | KL     | −176.996 | [−181.264, −173.012] |
| INT3         | flips  | −166.219 | [−171.214, −161.346] |

Table 11: Paired-difference bootstrap on MIXED − UNIFORM, all entries  $\times 10^{-3}$ ; negative favours MIXED. Both arms use seed = 42 over the same 128 sequence groups, so their replicates are the same resamples and the difference can be bootstrapped directly: 10,000 replicates, sequence-level resampling, rebuilt offline from the strided capture at one token in eight. Zero is excluded in all twelve cells, with 0.00% of replicates on the wrong side. Pairing narrows the interval by  $1.8\times$  to  $10.0\times$  relative to the sum of the two marginal interval widths. **This table is confirmatory, not the quoted interval:** the marginal intervals in Table 4 are measured on the full 32,768 routing tokens per layer, whereas this test can only be replayed on one token in eight, because per-token metric values were not retained. Trading the full sample for the pairing is not worth it for the headline, and no claim in the paper depends on the choice. Source: `results/<model>/paired_bootstrap.md`.

gation: OLMoE stays in 0.9968–0.9977 across every policy and bit-width against a GOLD of 0.9975, and Qwen in 0.9929–0.9996 against a GOLD of 0.9996, so a run whose worst layer starves 18 of 60 experts scores 0.9929.]

**[TODO:** Per-model figure sets are available as `olmoe_figures.pdf` and `qwen_figures.pdf` (9 pages each: perplexity, routing KL, top-1 flip rate, Jaccard distance, expert-usage entropy, output KL, then layer-wise routing drift at INT8, INT4 and INT3). Include the layer-wise pages if you have room — they show *where in depth* the drift accumulates, which nothing in the main text covers.]

## F Part 2: mechanism detail

## G Reproduction

**[TODO:** Short and concrete. Give the repository URL, the git commit recorded in the results files, the two commands that produce everything (`scripts/run.py`

| Run                                  | Bits | Unused | Worst $H$     | Pooled $H$ |
|--------------------------------------|------|--------|---------------|------------|
| <i>OLMoE, 16 layers, 1,024 slots</i> |      |        |               |            |
| GOLD                                 | BF16 | 0      | 0.9533        | 0.9975     |
| UNIFORM                              | INT4 | 0      | 0.9483        | 0.9976     |
| MIXED                                | INT4 | 0      | 0.9508        | 0.9976     |
| ATTENTION                            | INT4 | 0      | 0.9498        | 0.9976     |
| UNIFORM                              | INT3 | 0      | 0.9389        | 0.9968     |
| MIXED                                | INT3 | 0      | 0.9425        | 0.9973     |
| ATTENTION                            | INT3 | 0      | 0.9465        | 0.9974     |
| <i>Qwen, 24 layers, 1,440 slots</i>  |      |        |               |            |
| GOLD                                 | BF16 | 0      | 0.9742        | 0.9996     |
| UNIFORM                              | INT4 | 0      | 0.9703        | 0.9993     |
| MIXED                                | INT4 | 0      | 0.9694        | 0.9995     |
| ATTENTION                            | INT4 | 0      | 0.9694        | 0.9994     |
| UNIFORM                              | INT3 | 7      | <b>0.7585</b> | 0.9929     |
| MIXED                                | INT3 | 0      | 0.8752        | 0.9964     |
| ATTENTION                            | INT3 | 0      | 0.7850        | 0.9944     |

Table 12: Why the pooled expert-usage entropy reads as a null. **Pooled  $H$**  is the normalised entropy that earlier drafts reported: it concatenates every layer’s tokens into one histogram before measuring, so imbalances pointing in different directions across layers cancel. **Worst  $H$**  is the same quantity computed per layer, on the same runs. On Qwen UNIFORM INT3 the pooled number moves  $0.9996 \rightarrow 0.9929$  while the worst layer falls to 0.7585 and seven expert slots receive exactly zero tokens across five layers. **Unused** counts slots with zero selections at full resolution. ATTENTION partially mitigates the collapse as well — 74 starved slots against UNIFORM’s 222 and MIXED’s 55, with no fully unused expert — which is a further instance of the divergence in Section 6.3. INT8 rows are omitted: every policy is at the GOLD baseline there. Source: results/<model>/collapse.md.

| Run        | UNIFORM | MIXED  | PLACEBO | ATTENTION |
|------------|---------|--------|---------|-----------|
| OLMoE INT8 | 0.1445  | 0.1277 | 0.1445  | —         |
| OLMoE INT4 | 0.2362  | 0.1494 | 0.2362  | 0.2063    |
| OLMoE INT3 | 0.4341  | 0.2771 | 0.4341  | 0.3627    |
| Qwen INT8  | 0.2118  | 0.1853 | 0.2118  | —         |
| Qwen INT4  | 0.3558  | 0.2444 | 0.3558  | 0.3211    |
| Qwen INT3  | 0.7444  | 0.6166 | 0.7444  | 0.6346    |

Table 13: Top- $k$  Jaccard distance between the GOLD and candidate selected-expert sets, omitted from Table 3 for space. GOLD is 0.0000 by construction. It moves monotonically with the top-1 flip rate in every cell and shows the same ordering, including the ATTENTION rows sitting between MIXED and UNIFORM.

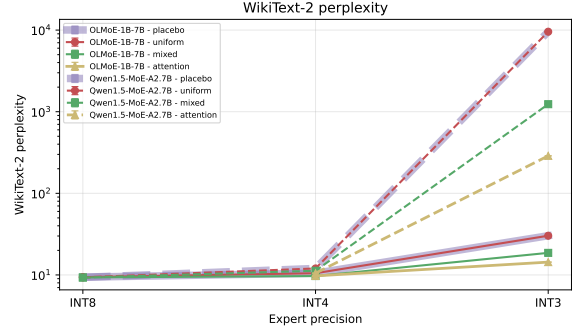


Figure 2: Cross-model detail panels. Page 1 of the source PDF is WikiText-2 perplexity; pages 2–4 are top- $k$  Jaccard distance, output-distribution KL over the gold top- $M$  support, and normalised expert-usage entropy. All four panels now include ATTENTION. [TODO: Select the page you want with [page=N]; page 1 is shown by default. Decide which one or two panels are worth the space and drop the rest. Note that page 1 duplicates the third panel of Figure 1, so page 2 or 3 is the better choice.]

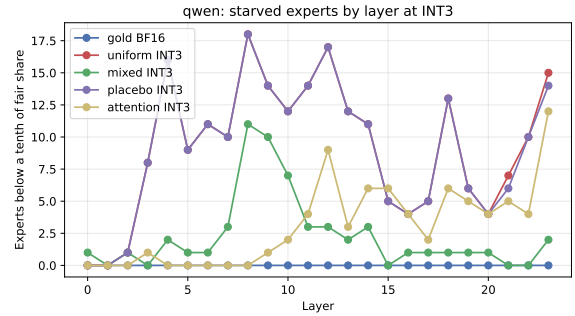


Figure 3: Starved experts by layer on Qwen at INT3, the one cell of the sweep where collapse occurs. Page 3 of the source PDF; pages 1–2 are the same panel at INT8 and INT4, pages 4–6 the per-layer usage entropy at the three bit-widths, and page 7 the expert-load concentration summary. [TODO: The OLMoE counterpart is collapse\_olmo.pdf and shows the null, which is worth including if there is room.]

|       |      | Routing KL vs. cell A |          |          |
|-------|------|-----------------------|----------|----------|
| Model | Bits | B                     | C        | D        |
| OLMoE | INT8 | 0.001833              | 0.007284 | 0.009110 |
|       | INT4 | 0.017479              | 0.009920 | 0.027223 |
|       | INT3 | 0.058610              | 0.039325 | 0.092915 |
| Qwen  | INT8 | 0.005292              | 0.020749 | 0.026038 |
|       | INT4 | 0.044367              | 0.038740 | 0.083575 |
|       | INT3 | 0.166601              | 0.314523 | 0.453984 |

Table 14: Routing KL by mechanism, showing the same INT8-to-INT4 crossover as the flip rates in Table 7 on a continuous metric:  $B < C$  at INT8 and  $B > C$  at INT4 on both models.

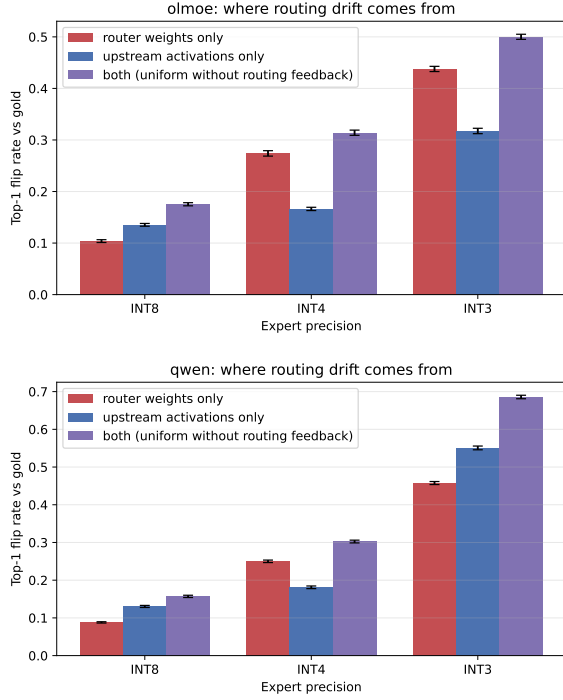


Figure 4: Top-1 flip rate for each mechanism at each precision: OLMoE (top) and Qwen (bottom). Error bars are 95% bootstrap intervals over sequences, 10,000 replicates. The crossover is the finding: at INT8 the activations-only bar exceeds the weights-only bar, and at INT4 the ordering reverses on both models. **[TODO: These source PDFs have four pages each — page 1 is this mechanism bar chart and pages 2–4 are the per-layer splits at INT8/INT4/INT3. includegraphics takes page 1 by default, which is what we want here; use [page=2], [page=3], [page=4] for the per-layer panels. Verify after the first compile that the intended panel appeared.]**

| Model | Bits | C     | MIXED | D     | UNIFORM |
|-------|------|-------|-------|-------|---------|
| OLMoE | INT8 | 13.54 | 13.76 | 17.54 | 18.01   |
|       | INT4 | 16.62 | 16.86 | 31.41 | 33.01   |
|       | INT3 | 31.74 | 32.15 | 50.01 | 52.52   |
| Qwen  | INT8 | 13.04 | 12.75 | 15.70 | 15.65   |
|       | INT4 | 18.11 | 17.90 | 30.25 | 30.62   |
|       | INT3 | 55.06 | 55.69 | 68.58 | 72.48   |

Table 15: Cross-check of the rebuilt cells against the Part 1 runs they should reproduce (top-1 flip rate, %). Cell C is the MIXED condition, and it tracks it to within a few tenths of a point — the residual being that Part 1 measures every token while the cells use the strided capture subset. Cell D is UNIFORM *without routing feedback*, so the D-versus-UNIFORM gap is not reconstruction error but a measurement of the feedback term: it grows with damage, from 0.47 points at OLMoE INT8 to 2.51 at OLMoE INT3 and 3.90 at Qwen INT3.

| Model | Mean   | Worst layer | Threshold |
|-------|--------|-------------|-----------|
| OLMoE | 99.66% | 99.46%      | 95%       |
| Qwen  | 99.37% | 98.49%      | 95%       |

Table 16: Reconstruction gate: top-1 agreement between cell A, rebuilt offline in fp32 on CPU from fp16-stored activations, and the routing decisions the BF16 GPU GOLD run recorded. Identical at every bit-width, since cell A depends only on gold quantities. Both models pass with substantial margin.

for the sweep, `scripts/analyze.py` and `scripts/attribute.py` for the tables and figures, and `scripts/compare_models.py` for the cross-model figure), the exact package versions from Section 5, and the GPU request lines. Mention that the test suite runs on CPU in a few seconds with no downloads, so the pipeline logic can be checked without cluster access.]